

APPLICATIONS SECTION

EFFECTS OF NOISE ON AMMI AND  
HIERARCHICAL CLASSIFICATION ANALYSES

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**Summary:** The Additive Main effects and Multiplicative Interaction (AMMI) model is used for the analysis of trials in which cultivars are tested at a different number of locations. The data used in this study originated from the South African National Lucerne Evaluation Program. The application of the AMMI model resulted in rankings of cultivars at different environments which could readily be explained by their breeding history and dormancy, while this was not obvious using the ANOVA model. The cultivar by environment means were then subjected to a hierarchical classification (HC) to group the cultivars into homogeneous groups. It was found that, in contrast to the results of HC, the AMMI biplot presents an agronomically interpretable pattern as far as the grouping of the cultivars and environments is concerned. In a simulation experiment conducted to determine the influence on pattern recovery under multiplicative noise, AMMI seemed to focus more on pattern than HC.

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1. INTRODUCTION

Any two-way design with potentially interpretable interaction may be analysed according to the Additive Main effects and Multiplicative

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Interaction (AMMI) statistical model. A body of literature has been developed which demonstrates the effectiveness of the AMMI model for understanding genotype-environment (GE) interactions in yields, estimating yields more accurately and selecting superior genotypes more reliably from multilocation yield trials, as well as allowing more flexible and efficient experimental designs (Kempton, 1984; Gauch & Zobel, 1988 & 1989; Crossa, Gauch & Zobel, 1990; Crossa, Fox, Pfeiffer, Rajaram & Gauch, 1991).

In the general statistical context the AMMI model was described by Williams as early as 1952. However, this is one of the methodologies which could only come into practical use with the advent of high-speed computers. The case of a multiplicative interaction matrix was treated by Williams (1952), Corsten & Van Eijsbergen (1972), Johnson & Graybill (1972) and Snee (1972). Gabriel (1978) gave a more rigorous and general treatment of the problem. In a non-rigorous way, Gollob (1968) introduced the FANOVA model which is equivalent to the AMMI.

The AMMI model incorporates both additive and multiplicative components into a least squares analysis. This model with prediction assessment was proposed for analysis of two-way tables obtained from cultivar trials in different localities and named the AMMI model by Gauch & Zobel (1988). Such an analysis of yield data from a cultivar trial is a useful extension of the more familiar analysis of variance (ANOVA), principal components analysis (PCA) and linear regression procedures, particularly when a large GE interaction is present.

By contrast, hierarchical cluster analysis techniques begin by examining close point pairs and small differences in distances between pairs of points. Attention is focused on the aspect of the data which is primarily either noise or is a consequence of sample density (Gauch, 1981). The yield obtained by a specific genotype in an environment is a function of both the genetic composition of the genotype and of the environmental differences. If a numerical classification of the genotypes (rows) over all environments (columns) is carried out, what is being done is to 'even out' the differences

between the environments by reducing them all to the same variance (or some equivalent standardization). Thus it is assumed that all differences reside in the genotype totals. Similarly, by the symmetric classification, all genotypes are brought to a common level of environments, and standardized by the rows, it is assumed that all the differences reside in the column (environmental) totals (Williams, 1976).

A two-way table analysis thus represents an attempt to improve these two arbitrary procedures. An AMMI analysis and its resulting biplot graph (genotypes or rows and environments or columns plotted on the same graph) of the two-way table (rows  $\times$  columns) is a dual analysis. This represents an attempt at improving the arbitrary procedure given above.

The objective of this research is to demonstrate the valuable characteristic of the AMMI model to focus on pattern, compared to other multivariate techniques, specifically hierarchical classification, which is used by plant breeders after ANOVA to group genotypes and environments. These may be more sensitive to noise than are grouping methods, such as PCA and AMMI.

## 2. NOISE REDUCTION BY EIGENVECTOR (PCA) ANALYSIS

Apposite multivariate models selectively recover pattern in their 'early' degrees of freedom and selectively recover noise in their 'late' degrees of freedom, i.e. the first few singular components and last few of the singular value decomposition (SVD) of a data matrix. Gauch (1982) used simulated data having both pattern and noise to demonstrate this selectivity for a SVD based technique called correspondence analysis (CA). Pattern involves sizeable correlations among matrix values, whereas noise involves deviations in individual matrix cells, therefore eigenanalysis (as in CA, PCA or AMMI) initially concentrates on the pattern (Gauch, 1988). After most of the pattern is exhausted, eigenanalysis at first extracts noise at a somewhat accelerated rate by exploiting chance correlations in the noise, but this is soon exhausted and the remainder of the sums of squares (SS) is extracted at

a slower rate. While pattern is processed in the early degrees of freedom, the extraction of noise is suppressed because the concentration on the pattern inhibits the exploitation of chance correlations in the noise.

The larger the dimensions of the matrix and the greater the noise, as in large GE trials, the greater will be this selectivity. Hence a large, noisy matrix of yield observations is ideal for demonstrating this selectivity. If the model is not even remotely apposite, then of course no selectivity can be expressed, while a more apposite model will exhibit greater selectivity (Gauch, 1988). The AMMI model's relationship to Taylor series expansions implies its adequate appropriateness and effectiveness in a wide variety of modelling situations (Bradu, 1984).

### 3. THE AMMI STATISTICAL MODEL

The AMMI model has a long history and is known under a variety of names, such as the multiplicative interaction (MI) model (Williams, 1952) and Gollob's (1968) FANOVA model. It is also described by authors Bradu & Gabriel (1978), Kempton (1984), Gauch (1988), Zobel, Wright & Gauch (1988), Gauch & Zobel (1989, 1990) (and others). The AMMI model can be described in the following manner.

Yield data  $Y_{ge}$  for  $G$  genotypes in  $E$  environments is considered, either unreplicated or as averages over  $R$  replications. Each genotype and environment combination, or more generically each row and column combination, is traditionally termed a 'treatment combination' or a 'cell', but for the sake of brevity the term 'treatment' will often be used here. In other words, the experiment has a two-way factorial design, with each treatment specified by a genotype and environment combination. The AMMI model is:

$$(1) \quad Y_{ge} = \mu + \alpha_g + \beta_e + \sum_{n=1}^N \lambda_n \gamma_{gn} \delta_{en} + \rho_{ge}$$

where  $\mu$  is the grand mean,  $\alpha_g$  the genotype additive effects (genotype deviations from the grand mean),  $\beta_e$  the environment additive effects,  $N$

the number of interaction PCA (IPCA) components retained in the AMMI model,  $\lambda_n$  the eigenvalue for IPCA component  $n$ ,  $\gamma_{gn}$  and  $\delta_{en}$  the genotype and environment eigenvectors for component  $n$ , and  $\rho_{ge}$  the residual. This model can be regarded as a 'cell means' model with one observation per cell, thus  $Y_{ge} = EY_{ge} + \rho_{ge}$ , where a rather sophisticated 'additive cum multiplicative' model was assumed for the cell means. Now, without loss of generality, one can assume that  $\Sigma\alpha = \Sigma\beta = 0$ ; in addition,  $\Sigma\gamma = \Sigma\delta = 0$ . The  $\gamma$  and  $\delta$  eigenvector values for each IPCA are scaled to unit vectors so that  $\Sigma\gamma^2 = \Sigma\delta^2 = 1$ . The eigenvalue for a given IPCA component equals  $\lambda^2$ . The sum of the eigenvalues  $\Sigma\lambda^2$  for  $N$  components, plus the residual SS of  $\Sigma\rho^2$  for a reduced model, equals the GE interaction SS.

The multiplicative part  $\Sigma\lambda_n \gamma_{gn} \delta_{en}$  is a sum of rank one terms; each such term can be written in an infinity of ways as the product of two vector factors, where one is a genotype score vector and the other an environment score vector. In forming these two vectors, one can (i) assign the  $\lambda$  to the  $\gamma$ , which favours the genotypes as far as Euclidean representation is concerned, (ii) assign the  $\lambda$  to the  $\delta$ , which likewise favours the environments, (iii) assign equally roots  $\lambda$  to  $\gamma$  and to  $\delta$  which gives the factorization used by Bradu & Gabriel (1978) for model diagnosis. In the latter case both scores are in units of the yield square root and hence the product of a genotype score ( $\lambda^{0.5}\gamma_g$ ) times an environment score ( $\lambda^{0.5}\delta_e$ ) is in the units of yield as required. Thus multiplication of a genotype score by an environment score gives the estimated interaction directly, without need of a further multiplication by  $\lambda$ .

When a cultivar and an environment have a score with the same sign, their interaction is positive; if different, their interaction is negative. If a cultivar or an environment has a score of nearly 0 it has small interaction effects (and hence can be fitted well by an additive model).

If the experiment is replicated, the individual observation  $Y_{ger}$  for replicate  $r$  may be modelled by adding to the above equation an error term  $\epsilon_{ger}$  which equals  $Y_{ger}$  minus the  $Y_{ge}$  mean, thus  $Y_{ger} = EY_{ge} + \epsilon_{ger}$ .

Component  $n$  then has an eigenvalue of  $R\lambda_n^2$ . In this form, we have the general 'cell means' model, with an arbitrary number of replications per cell. If scaling of the additive effects as means is preferred to deviations, as in the above AMMI model (1), then the means  $\mu + \alpha_g$  and  $\mu + \beta_e$  are used instead of the deviations  $\alpha_g$  and  $\beta_e$  and the AMMI model (1) modified to subtract rather than add the grand mean  $\mu$ .

#### 4. PARAMETER ESTIMATION AND ANOVA IN THE AMMI MODEL

In the context of yield trials, retention of the original yield or observational units is preferred to the standard statistical practice of attempting to find a suitable transformation to remove or reduce the interaction (Cox, 1984; Gauch, 1988). Typically, the logarithmic transformation is used to improve the statistical properties of the data. But the logarithmic transformation increases the influence of the small yield values at the expense of the large yields, in direct opposition to the usual agronomic perspective which focuses interest on the large yields. Thus, least squares analysis on the untransformed data is the natural approach for experimental data such as in agriculture, where interpretation on the original scale of measurement is preferred. Transformation should only be applied to bring the data closer to homoscedasticity and normality rather than to make them additive (Bradu, 1984).

For balanced data, the least squares fit is obtained by fitting the additive part of the AMMI model ( $\mu$ ,  $\alpha_g$  and  $\beta_e$ ) with ordinary ANOVA and analysing the nonadditive residual or interaction by fitting the multiplicative part ( $\lambda_n$ ,  $\gamma_{gn}$  and  $\delta_{en}$ ) with PCA (Gabriel, 1978).

In the general AMMI model, the interaction components' contributions are given as squared singular values. Gollob (1968) ascribed to them a simple method of calculating degrees of freedom and Mandel (1969) attempted an ANOVA where the degrees of freedom for each squared singular value is determined by Monte-Carlo methods. The smallest, non-significant ones are pooled into a variance SS (named 'residual' in the AMMI analysis) and all the rest can be tested in an ANOVA once a decision

has been made on how many components to retain. Gauch (1988) reasoned that for experimental data the adjustment of the degrees of freedom should be in the opposite direction to that of Mandel (1969). Hence, for the early components of greatest interest and for data with pattern and noise, Gollob's (1968) simple method to calculate the  $n$ -th PCA component degrees of freedom ( $df = G + E - 1 - 2n$ ) would be preferable. A more correct procedure to obtain the interaction components' degrees of freedom for experimental data with real main and interaction effects is suggested by H.G. Gauch (unpublished: Degrees of freedom in the Mandel model for two-way tables).

The multiplicative term of the AMMI model is  $\sum_n \lambda_n \gamma_{gn} \delta_{en}$ . Ordinarily the number  $N$  of IPCA components retained in the model are chosen with empirical considerations of F-tests of significance, predictive accuracy, agricultural interpretability of the associated IPCA scores, etc. Usually  $N$  is 0-3 and most frequently 1, thus producing a reduced model with a residual which combines the discarded components. Therefore, for  $N = 1$ , the multiplicative term is partitioned as

$$\sum_{n=1}^N \lambda_n \gamma_{gn} \delta_{en} = \lambda_1 \gamma_{g1} \delta_{e1} + \sum_{n=2}^N \lambda_n \gamma_{gn} \delta_{en},$$

where the last term is the discarded residual. In the AMMI with  $N=1$ , the interaction scores unravel the true elements of the structure (Bradu, 1984). In fitting a more sophisticated model, one has to include in every cell's estimate the contribution of all the observations, as opposed to the cell means model, where each estimate is based only on the cell's values.

As a result of the discarded residual the AMMI model does not fit the data perfectly and hence is biased, whereas the cell means model is unbiased. The accuracy of a yield estimate depends upon both the variance of the estimate (or precision) and the magnitude of bias. The 'Stein effect' (Stein, 1955; James & Stein, 1960) has been observed with the AMMI model, i.e. that the gain from the reduction of variance resulting from the use of a biased estimator sometimes more than offsets the inaccuracy due to bias

(Gauch, 1988; Gauch & Zobel, 1988, 1989, 1990). The most obvious way for increasing the accuracy of a yield estimate is to use an increased number of replications  $R$ , because the standard error of a cell mean decreases with the square root of  $R$ . However, there are practical limitations and, furthermore, the square-root dependency implies diminishing returns. Hence alternative strategies are important, whether at the experimental level, such as blocking, or at the modelling and analysis stage, such as the use of the AMMI model.

## 5. APPLICATION OF THE AMMI ANALYSIS TO AGRICULTURAL DATA

The data to which the AMMI model was applied originated from the South African National Lucerne Evaluation Program. This programme was designed by Dr. A. Smith of the Roodeplaat Grassland Institute to give an indication of the suitability of different lucerne cultivars to environmental and agricultural conditions in South Africa. A series of 24 trials in both the Summer and Winter Rainfall Regions and covering both dryland and irrigated conditions were established as randomised blocks with sixteen cultivars in three replicates at each site.

The lucerne cultivars were representative of dormancy groups ranging from a dormancy of 6 to 9 on the international scale and covering semi-dormant, intermediate, non-dormant and highly non-dormant groups (Hanson, *et al.*, 1988). Dormancy is a genetically controlled factor which acts as an escape mechanism for unfavourable or stress conditions such as cold, pests and diseases, moisture stress (excess as well as deficiency) and grazing pressure. These stress factors seem to be additive and it is expected that the greater the accumulative effect the better the performance of the more dormant cultivars should be (A. Smith, pers. comm.). Conversely, it is expected that highly non-dormant cultivars should perform best under favourable conditions. Using the usual additive model (ANOVA), these expected trends were not obvious, but the application of the AMMI model resulted in rankings of cultivars that could be readily explained by the

breeding history and dormancy of the different cultivars (Smith, 1992).

### 5.1 The AMMI analysis of the data

MATMODEL Version 2.0 (Gauch, 1990a) allows each observation to be marked for modelling or validation of the within-trial accuracy by measuring predictive success with a simple data-splitting method. The ability of a model to predict validation data not used in constructing the model constitutes predictive success (Gauch, 1988, 1990b). The validation of the AMMI model for this data set indicated the AMMI1 model, i.e. partitioning the GE interaction into the first IPCA (IPCA 1) component and a nonsignificant residual, as the best predictive model. Finally, AMMI1 estimates for all cultivar by environment combinations were generated using this model with three replicates.

The partitioning of the total 'treatment' (TRT) SS into cultivar, environment (24 sites by 2 years) and interaction are 0,2%, 97,8% and 2,0% of the TRT SS, respectively (table 1). The interaction SS is ten times larger than the cultivar SS. The IPCA 1 component alone explains 54,4% of the interaction SS in only 8,7% of the interaction degrees of freedom and has a SS that is 5,1 times larger than the cultivar SS. The interaction pattern in the IPCA 1 scores is related to dormancy ratings for the cultivars (fig. 1) and to moisture availability of the environments, and therefore has a meaningful agricultural interpretation.

Table 1. AMMI analysis of variance for lucerne yields

| Source   | df   | SS     | MS      | Probability |
|----------|------|--------|---------|-------------|
| Total    | 2303 | 124303 | 53,97   |             |
| TRT      | 767  | 121868 | 158,89  | < 0,001     |
| CULT     | 15   | 259    | 17,30   | < 0,001     |
| ENV      | 47   | 119179 | 2535,72 | < 0,001     |
| C × E    | 705  | 2429   | 3,446   | < 0,001     |
| IPCA1    | 61   | 1322   | 21,67   | < 0,001     |
| Residual | 644  | 1108   | 1,720   | 0,1065      |
| Error    | 1536 | 2435   | 1,585   |             |

The AMMI1 model partitions the TRT SS and degrees of freedom following a model which contains 99,1% of the TRT SS. Although a very small residual is discarded, i.e. only 0,9% of the TRT SS, it is important as it is this discarding that causes AMMI estimates to differ from mere cell means, and although biased, to have potential for greater predictive success (Gauch, 1988, 1990b; Gauch & Zobel, 1988). As this discarded residual is not negligible, these adjustments change the cultivar rankings within each environment considerably.

## 5.2 Description of the lucerne cultivars tested in the trial

Although cultivar SS has a dormancy rating of 7, it is consistently outperformed by cultivars of similar dormancy (table 2). It is inherently a weak hay producer and therefore does not obtain a high ranking, even under stress conditions. It has good grazing tolerance and the AMMI rankings lead us to believe that the dormancy rating should be much lower, similar to

**Table 2. Dormancy ratings of the sixteen lucerne cultivars tested in the AMMI analysis**

| Number | Cultivar    | Designation | Rating | Origin |
|--------|-------------|-------------|--------|--------|
| 1      | SA STANDARD | SS          | 7      | RSA    |
| 2      | GRANADA     | GR          | 9      | USA    |
| 3      | CUF 101     | CU          | 9      | USA    |
| 4      | BARONET     | BA          | 7      | USA    |
| 5      | PIERCE      | PI          | 8      | USA    |
| 6      | SAPHIRE     | SA          | 8      | USA    |
| 7      | TOPAZ       | GT          | 8      | USA    |
| 8      | METEOR      | CH          | 6      | USA    |
| 9      | RESISTO     | RE          | 8      | USA    |
| 10     | SIRIVER     | SI          | 8      | AUS    |
| 11     | TRIFECTA    | TR          | 8      | AUS    |
| 12     | WL 515      | W5          | 8      | USA    |
| 13     | WL 516      | W6          | 8      | USA    |
| 14     | WL 605      | W0          | 9      | USA    |
| 15     | WL 517      | W7          | 8      | USA    |
| 16     | DIAMOND     | DI          | 7      | USA    |

those of other grazing cultivars (Smith, 1992). Cultivars RE and GT are a duplicate entry and should therefore give similar results. Cultivars SI and TR are Australian-bred cultivars and do not seem to have the same yield potential as the USA cultivars; they react as if they ought to have a dormancy rating of one unit lower than that indicated.

### 5.3 Interpretation of the AMMI scores

In figure 1 the cultivars with lower dormancy ratings (generally associated with low yields) group to the left and the cultivars with higher dormancy ratings (associated with higher yields generally) to the right of the graph. The response to environmental conditions showed a high correlation to this dormancy gradient, a fact not generally appreciated by pasture advisors.

**Table 3. The cultivar means, dormancy ratings and first interaction PCA component scores of the AMMI1 analysis, ranked in descending order of the cultivar scores**

| Number | Name | Dormancy | Mean  | Score  |
|--------|------|----------|-------|--------|
| 8      | CH   | 6        | 8,734 | 2,455  |
| 11     | TR   | 8        | 9,169 | 1,842  |
| 4      | BA   | 7        | 9,539 | 1,168  |
| 16     | DI   | 7        | 9,762 | 0,975  |
| 10     | SI   | 8        | 9,197 | 0,537  |
| 1      | SS   | 7        | 8,919 | 0,294  |
| 2      | GR   | 9        | 9,701 | 0,272  |
| 6      | SA   | 8        | 9,838 | -0,008 |
| 12     | W5   | 8        | 9,514 | -0,234 |
| 14     | W0   | 9        | 9,732 | -0,560 |
| 3      | CU   | 9        | 9,761 | -0,638 |
| 5      | PI   | 8        | 9,744 | -0,746 |
| 13     | W6   | 8        | 9,887 | -0,987 |
| 9      | RE   | 8        | 9,526 | -1,175 |
| 7      | GT   | 8        | 9,671 | -1,516 |
| 15     | W7   | 8        | 9,827 | -1,680 |

Figure 1 displays both cultivar main effects and interaction. For example, cultivars SS and GR differ in main effect but not interaction (IPCA 1 score), whereas cultivar GR differs from cultivar GT only in interaction, since the

scores differ but the mean yields do not. Thus it can be seen that cultivars SS and GT differ both in main effect and interaction.

A biplot (Bradu & Gabriel, 1978; Kempton, 1984) of the AMMI1 results where the abscissa shows the main effects (cultivar and environment mean yields) and the ordinate shows the IPCA 1 scores, gives further insight into the data (Smith, 1992).

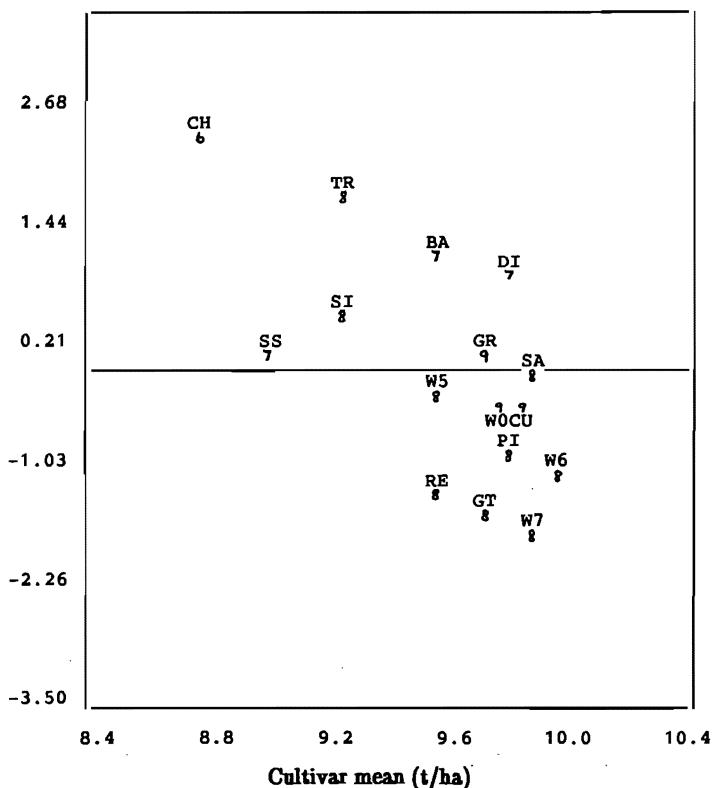


Figure 1. Plot of the AMMI1 cultivar first interaction PCA component scores versus the cultivar mean yields, also indicating dormancy ratings

## 6. NUMERICAL CLASSIFICATION OF AGRICULTURAL DATA

Numerical algorithms, e.g. cluster analysis, to classify data are widely used and respected in the general scientific community (ecologists, agronomists and others), in particular algorithms of hierarchical classification (HC).

Cluster analysis requires a measure of similarity between the individuals to be classified and it imposes a discontinuity in the data. The method has been widely used to study genotype adaptation by simplifying the pattern of responses and to subdivide genotypes and environments into more homogeneous groups. Comprehensive reviews of the application of cluster analysis to the study of GE interactions can be found in Lin, *et al.* (1986) and Westcott (1987).

Some of the disadvantages of cluster analysis are: (i) numerous hierarchical grouping algorithms exist and each of them may produce different cluster groups; (ii) the truncation level of the classificatory hierarchies may be decided arbitrarily; (iii) many different similarity measures can be used yielding different results; and (iv) cluster analysis may produce misleading results by showing structures and patterns in the data when they do not exist.

### 6.1 Hierarchical cluster analysis of the data

The cultivar-by-environment means from the original data set were used to perform an HC, to group the 16 cultivars (by 48 environments) into homogeneous groups by the UPGMA technique using agglomerative hierarchical cluster analysis and the group average criterion for forming clusters (Genstat 5, 1987). The unweighted pair-group method using arithmetic averages (UPGMA) is a polythetic agglomerative technique (Sneath & Sokal, 1973, p. 230). They consider UPGMA the most frequently used clustering technique and recommend its use whenever there is no specific reason for choosing some other technique.

The last two clusters to be formed in figure 2(a) contain cultivars SS and CH (average yields  $< 9$  tonnes  $\text{ha}^{-1}$  and lowest dormancy ratings of 6 and 7) in one cluster and the rest of the cultivars in another cluster (average

yields > 9 tonnes ha<sup>-1</sup> and higher dormancy ratings of 7 to 9). Yields and dormancy ratings are given in table 3 and also illustrated in figure 1.

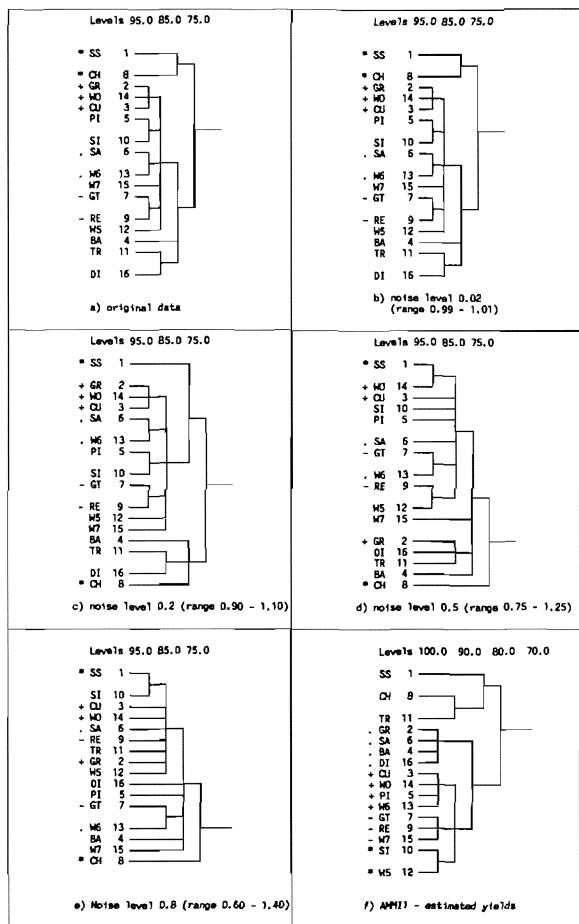


Figure 2. Hierarchical clustering dendrogrammes of the cultivar by environment mean yields of the lucerne data

## 6.2 Comparison of AMMI plots and HC clusters

Cultivars RE and GT, which are a duplicate entry, yielded 9,526 and 9,671 tonnes  $\text{ha}^{-1}$  and have scores of -1,17 and -1,52, respectively (table 3). Thus they take up very close positions in figure 1, as expected. Cultivars SI and TR react as if they should have a dormancy rating of one unit lower than that indicated.

However, in the HC clustering, the similarity of cultivars GT and RE (figure 2(a), marked -) is expected as they are a duplicate entry. The Australian-bred cultivars SI and TR, with the same dormancy rating and yield potential (table 3) were expected to be similar and indeed they group together (figure 1). However, SI is grouped with PI (which has a higher yield potential) at the first stage. Cultivars SA and W6 (marked ·) both have dormancy ratings of 8 and similar yields. GR, W0 and CU (marked +) are the cultivars with the highest dormancy rating of 9 and similar yields.

Thus, in opposition to the results of HC, the AMMI plot (figure 1) *presents an agronomically interpretable pattern as far as the grouping of the cultivars is concerned.*

## 7. INCORPORATING NOISE INTO THE ORIGINAL DATA

In a simulation experiment to gain insight into pattern recovery under multiplicative noise of various magnitudes, four sets of pure noise were generated using values taken from a uniform distribution in the ranges 0,99 – 1,01, 0,90 – 1,10, 0,75 – 1,25 and 0,60 – 1,40. Each set of noise was multiplied with the original data to incorporate increasing levels of noise into the data. Thereafter each data set was subjected to ANOVA, AMMI and HC analyses, respectively. The main effect analysis (ANOVA) coefficient of variation increases from 11,0% to 31,1% when successively increasing the noise level (table 4). The grand means are virtually unaltered, but the *discarded residual* of the AMMI1 analyses increases from about 1% to more than 4% of the treatment SS.

**Table 4. Summary of results after application of main effects ANOVA, AMMI and HC analyses on data containing different levels of noise**

| Noise range | Noise level | Grand mean | ANOVA CV % | AMMI residual SS as a % of TRT SS |
|-------------|-------------|------------|------------|-----------------------------------|
| none        | 0           | 9,533      | 11,0       | 0,909                             |
| 0,99 – 1,01 | 0,02        | 9,531      | 11,0       | 0,911                             |
| 0,90 – 1,10 | 0,2         | 9,553      | 13,5       | 1,030                             |
| 0,75 – 1,25 | 0,5         | 9,645      | 21,0       | 2,090                             |
| 0,60 – 1,40 | 0,8         | 9,500      | 31,1       | 4,410                             |

The perturbation of the data should be seen in the following perspective. The scale of the original data is tonnes  $\text{ha}^{-1}$ . Incorporating a noise level of 0,8 (range 0,60 – 1,40) therefore implies an increase or decrease of the data by up to 0,4 tonnes  $\text{ha}^{-1}$ . This is a deviation of 0,4 tonnes  $\text{ha}^{-1}$  of the grand mean of 9,533 tonnes  $\text{ha}^{-1}$ , which is derived from a data set with a range of 0,432 to 35,273, and therefore represents 4,2% of the grand mean.

#### **7.1 Comparison of the AMMI plots and HC dendrograms under increasing noise**

The dendrograms from the HC analyses on the cultivars, as is figure 2(a), but with the four levels of noise incorporated into the original data show that cultivars GT and RE (marked -) would be expected to be grouped together as they are a duplicate entry (figure 2(b) to 2(e)). Furthermore, it is expected that cultivars SS and CH (marked \*), which are the cultivars with the lowest dormancy ratings (grazing types) and average yields, as well as cultivars W0, GR and CU (marked +) and cultivars SA and W6 (marked .), which have similar high ratings (9 and 8, respectively) and high average yields should group together because of similarities (table 3).

These dendrograms illustrate the following shifts in patterns: (i) at noise level 0,02 there is no apparent difference from the original clustering (figure 2(b)); (ii) when incorporating a level of 0,2 (figure 2(c)), an expected

similarity is seriously disturbed, that is, cultivars SS and CH, move to opposite ends of the dendrogram; (iii) at 0,5 noise level (figure 2(d)), more serious disturbances are introduced which are exaggerated at the 0,8 level (figure 2(e)). None of the expected similarities as depicted by the marked groupings (\*, +, · and -) are now obtained. This indicates how unreliable the groupings by the HC analysis will become as the noisiness of data increases.

Table 5. AMMI1 scores for the original data and the four levels of incorporated noise

| Noise level |      | 0,0   | 0,02  | 0,2   | 0,5   | 0,8   |
|-------------|------|-------|-------|-------|-------|-------|
| Number      | Name |       |       |       |       |       |
| 8           | CH   | 2,45  | 2,46  | 2,23  | 2,51  | 3,27  |
| 11          | TR   | 1,84  | 1,85  | 1,99  | 2,00  | 1,57  |
| 4           | BA   | 1,17  | 1,16  | 1,24  | 1,76  | -0,84 |
| 16          | DI   | 0,97  | 0,99  | 1,05  | 0,51  | 0,61  |
| 10          | SI   | 0,54  | 0,52  | 0,41  | 0,51  | -0,39 |
| 1           | SS   | 0,29  | 0,28  | 0,09  | 0,58  | 0,19  |
| 2           | GR   | 0,27  | 0,28  | 0,57  | 0,40  | -1,30 |
| 6           | SA   | -0,01 | -0,01 | 0,09  | -0,60 | 0,41  |
| 12          | W5   | -0,23 | -0,24 | -0,37 | -0,93 | 1,07  |
| 14          | W0   | -0,56 | -0,55 | -0,46 | -0,12 | -0,83 |
| 3           | CU   | -0,64 | -0,66 | -0,47 | -0,92 | -1,36 |
| 5           | PI   | -0,75 | -0,71 | -0,65 | -0,18 | -0,91 |
| 13          | W6   | -0,99 | -0,99 | -1,13 | -0,80 | -0,43 |
| 9           | RE   | -1,17 | -1,19 | -1,24 | -1,28 | 0,45  |
| 7           | GT   | -1,52 | -1,55 | -1,34 | -1,95 | -2,01 |
| 15          | W7   | -1,68 | -1,64 | -2,01 | -1,49 | 0,51  |

Compared to the previous analyses, the scores of cultivars with incorporated noise levels of 0,02 and 0,2 obtained from the AMMI1 analyses are similar to that of the original data (table 5). This can be seen more clearly from the AMMI1 cultivar IPCA 1 scores, presented in table 5 and the simple correlation coefficients between the scores of the original data set and perturbed data sets (table 6), which hardly change with the incorporation of noise levels of 0,02 and 0,2.

**Table 6.** Correlation coefficients between the AMMI first interaction PCA scores of the original data and data with incorporated noise

| Noise range | Noise level | 16 cultivars | 48 environments |
|-------------|-------------|--------------|-----------------|
| 0,99 - 1,01 | 0,02        | 1,000        | 1,000           |
| 0,90 - 1,10 | 0,2         | 0,989        | 0,989           |
| 0,75 - 1,25 | 0,5         | 0,951        | 0,961           |
| 0,60 - 1,40 | 0,8         | 0,600        | 0,810           |

There is evidence of a shift in pattern taking place at noise level 0,5, but this still does not seriously change the expected/original groupings. The correlation coefficient with the original scores is 0,951 (table 6). Cultivars GT and RE, as well as cultivars GR, W0 and CU seem to have larger differences in interaction than for the original data, and the latter show a slight difference in main effect between them. The cultivar which is influenced most is W5, having the largest change in interaction, from -0,23 to -0,93 at the 0,5 noise level (table 5).

Only at the noise level of 0,8 does the cultivar pattern become seriously affected. The correlation coefficient between the scores of the original data and this data set is now 0,600. Most noticeable, that cultivars GT and RE are not grouped together anymore (they now display a large interaction). Cultivars GR, W0 and CU, which are very similar in average yield and interaction, now appear different in main effect (average yield). The scores of cultivars RE, BA, GR, W7, W5 and SI change sign, the largest changes being those of BA from 1,17 to -0,84 and W7 from -1,68 to 0,51. These changes will affect the estimates of cultivar  $\times$  environment interaction means. It is interesting that in both the cases of cultivars and environments, the correlation coefficients between the scores of the original data set and the perturbed data sets, only become less than 0,95 at the noise level of 0,8 (table 6).

The HC analysis on the AMMI1-estimated values, on the other hand, produced the dendrogram illustrated in figure 2(f). These groupings are

more in accordance with the dormancy ratings and expected responses of the cultivars (figure 1 & table 3). For example, W0, CU, PI & W6 (marked +); RE, GT & W7 (marked -), as well as BA, DI, GR & SA (marked ·) form expected groups.

This simulation points to the likely conclusion that AMMI analysis focuses more on pattern, and is less influenced by noise, than hierarchical classification.

## 8. CONCLUSION

The value of a method which enables an agronomic enquiry to focus on pattern and to discard noise is also shown in the work of Nachit, *et al.* (1992) and Crossa *et al.* (1991). The latter investigated procedures for improving the rate of predictive success of a yield trial, grouping environments and genotypes into homogeneous subsets and determining the yield stability of 18 bread wheats evaluated at 25 locations. Locations were clustered using the AMMI1-estimated values of the 18 genotypes as attributes and, conversely, the genotypes were grouped using the AMMI1-estimated values of 25 locations as attributes. They found that groups of genotypes and locations were more cohesive after using AMMI1 estimates as input in the cluster analysis, as this led to clearer explanations of response patterns. Their conclusion was that integration of the AMMI analysis and HC on estimates with reduced noise was useful for interrelating the different objectives of multilocation trials: (i) to assess the success of yield prediction; (ii) to group sites for breeding; and (iii) to interpret genotype  $\times$  environment interaction.

This is in strong support of the conclusion reached in this study, which was made on the basis of a simulation study and illustrates the principle.

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